

ORSHA, Belarus

WMO: 267630

Lat: 54.503N

Lon: 30.444E

Elev: 607

StdP: 14.38

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-2.1	-12.2	2.9	-7.8	-6.4	4.0	-1.0	19.9	29.0	18.3	26.5	3.0	0	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	83.8	67.6	80.2	65.7	77.2	64.6	69.7	79.8	67.8	77.6	66.1	74.5	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	98.0	74.7	64.2	92.1	72.5	62.7	87.3	70.7	34.0	79.9	32.5	77.8	31.1	74.2	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.0	16.0	14.1	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			WB	-15.6	88.2	4.7	4.1	-19.0	91.2	-21.8	93.5	-24.5	95.8	-27.9	98.8
				-15.9	72.3	4.7	2.8	-19.3	74.3	-22.1	75.9	-24.7	77.4	-28.2	79.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	21.8	22.4	31.0	44.7	54.5	60.9	65.5	62.8	53.0	42.8	33.5	25.6	(6)
(7)		DBStd	17.57	11.26	11.82	8.60	7.80	7.71	6.04	5.58	6.08	6.50	7.27	8.76	10.72	(7)
(8)		HDD50	4013	873	772	590	196	41	1	0	0	44	243	496	757	(8)
(9)		HDD65	8075	1338	1192	1055	609	334	151	63	115	362	689	945	1222	(9)
(10)		CDD50	1578	0	0	1	37	181	329	479	397	134	20	0	0	(10)
(11)		CDD65	164	0	0	0	0	10	29	77	47	1	0	0	0	(11)
(12)		CDH74	1457	0	0	0	5	95	218	676	439	24	0	0	0	(12)
(13)		CDH80	339	0	0	0	0	12	23	169	134	1	0	0	0	(13)
(14)	Wind	WSAvg	6.4	7.2	7.1	7.0	6.4	6.2	5.8	4.9	5.1	5.5	6.6	7.4	7.4	(14)
(15)	Precipitation	PrecAvg	25.9	1.6	1.5	1.5	1.6	2.7	2.9	3.2	2.8	2.4	2.3	1.9	1.5	(15)
(16)		PrecMax	34.1	3.7	2.7	3.2	3.9	5.0	6.6	7.9	9.3	5.4	5.6	4.3	2.8	(16)
(17)		PrecMin	19.3	0.3	0.7	0.3	0.5	0.7	1.1	0.3	0.1	0.6	0.3	0.1	0.6	(17)
(18)		PrecStd	3.9	0.7	0.5	0.7	0.8	1.2	1.2	1.8	1.8	1.3	1.4	0.9	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	43.1	56.5	72.8	81.9	83.6	88.7	92.1	77.4	67.1	53.1	47.2	(19)
(20)		MCWB	40.6	39.9	46.9	54.9	66.2	66.2	69.3	69.3	62.2	55.3	51.2	45.2	(20)	
(21)		2%	DB	38.6	39.3	50.2	67.5	76.4	80.0	84.8	82.4	72.0	59.6	49.4	43.3	(21)
(22)		MCWB	36.9	37.0	42.5	53.3	63.7	65.6	69.0	65.9	60.4	54.4	47.4	41.8	(22)	
(23)		5%	DB	36.3	36.9	45.5	62.5	72.1	76.8	80.9	78.5	67.3	56.0	46.3	40.1	(23)
(24)		MCWB	35.1	35.2	40.0	51.1	59.4	64.6	67.2	64.3	57.7	51.9	44.4	39.1	(24)	
(25)		10%	DB	34.3	35.1	41.7	58.4	68.5	73.3	77.6	74.6	64.0	53.0	44.2	37.0	(25)
(26)		MCWB	33.2	33.7	37.9	48.8	57.4	62.6	65.5	62.9	55.9	50.0	42.8	35.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	40.5	47.8	57.5	68.5	70.0	73.1	70.8	63.9	58.3	51.6	45.7	(27)
(28)		MDCB	42.2	42.3	54.9	69.5	78.9	78.3	84.3	85.2	73.6	63.7	52.9	46.8	(28)	
(29)		2%	WB	37.2	37.2	43.5	54.4	64.4	67.9	70.6	68.4	61.0	55.2	47.8	41.9	(29)
(30)		MDCB	38.3	38.5	48.6	63.8	74.4	77.1	81.2	79.9	68.5	58.3	49.0	42.9	(30)	
(31)		5%	WB	35.1	35.5	40.5	52.2	61.8	66.1	68.4	66.6	59.0	52.4	44.8	39.0	(31)
(32)		MDCB	36.1	36.6	44.3	60.4	70.3	74.3	78.0	75.3	65.4	55.3	46.0	40.2	(32)	
(33)		10%	WB	33.5	34.0	38.2	50.0	58.9	64.0	66.6	64.5	57.1	50.2	42.9	35.8	(33)
(34)		MDCB	34.3	34.9	41.5	57.3	65.9	71.2	75.2	71.2	62.2	53.1	44.1	36.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.0	9.7	11.7	16.4	17.7	16.7	17.8	18.0	16.6	11.3	6.9	7.0	(35)	
(36)		5% DB	MCDBR	6.2	7.3	16.5	24.1	23.5	21.7	22.5	24.6	22.1	16.1	8.9	6.7	(36)
(37)		MCWBR	6.0	6.4	10.6	12.4	12.0	10.5	10.0	10.3	12.0	11.1	7.6	6.2	(37)	
(38)		5% WB	MCDBR	5.9	6.8	14.4	20.6	21.1	18.6	20.4	19.5	18.9	13.9	8.0	6.6	(38)
(39)		MCWBR	5.9	6.3	9.9	11.3	11.8	10.2	10.0	9.2	11.1	10.4	7.3	6.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.322	0.351	0.408	0.396	0.378	0.406	0.416	0.373	0.349	0.328	0.290	(40)	
(41)		taud	2.135	2.086	2.124	2.171	2.287	2.360	2.299	2.264	2.351	2.382	2.247	2.161	(41)	
(42)		Ebn at Noon	201	233	257	258	269	276	265	253	250	228	185	173	(42)	
(43)		Edh at Noon	21	31	37	41	39	36	38	37	30	23	19	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	210	496	938	1270	1663	1811	1735	1407	934	453	164	120	(44)	
(45)		RadStd	48	89	139	148	142	165	163	150	77	91	32	35	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +1.93	(j) N/A	(k) N/A	(l) N/A	(46)	
(47)	Regional (9 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	-313	+176	N/A	(47)

Nomenclature: See separate page